

Multi-Agent Vietnamese Summarization with Grade-Aware Difficulty Control for Primary Students

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Abstract — This paper proposes MAS-ViSum, a multi-agent framework for adaptive Vietnamese text summarization tailored for primary school students (Grades 1–5). The system integrates extractive (VES) and abstractive (VAS) summarization through specialized agents for planning, generation, evaluation, and orchestration. A core innovation is the dual-level difficulty control mechanism that dynamically adjusts linguistic complexity according to target grade and difficulty mode (Easy, Medium, Hard). Trained on over 10,000 educational texts and 22,000 summary pairs, MAS-ViSum achieves competitive ROUGE and BERTScore performance while improving readability and educational suitability compared to conventional baselines. Notably, VAS attains a ROUGE-1 of 0.82 and BERTScore of 0.76. Preliminary human evaluation by an experienced teacher confirms the summaries are clear, coherent, and suitable for classroom use. MAS-ViSum demonstrates the effectiveness of combining multi-agent architectures with grade-aware readability control for personalized educational applications in low-resource languages.

Keywords: Text summarization, Multi-agent systems, Readability control, Vietnamese NLP, Educational AI, Primary education

I. INTRODUCTION

In the era of digital information explosion, accessing and processing large volumes of text has become a significant challenge for primary school students, whose reading comprehension and information synthesis abilities remain limited. Automatic text summarization helps learners identify main ideas, eliminate redundant information, and develop logical thinking and knowledge retention skills [1, 2]. In Natural Language Processing (NLP), deep learning and large language models have substantially improved summarization systems, especially abstractive approaches. However, most existing systems follow a fixed “one-size-fits-all” design, making it difficult to control summary length, linguistic complexity, and difficulty level for specific user groups, particularly young learners [3, 4].

Multi-agent systems (MAS) have recently emerged as a promising solution by coordinating specialized agents for content generation, evaluation, and refinement, thereby enhancing coherence and adaptability [5, 6]. Nevertheless, current research mainly focuses on general-domain quality and

has paid limited attention to grade-specific readability control in primary education, especially for low-resource languages like Vietnamese. Vietnamese presents unique challenges, including variable sentence structures, vocabulary density, and heavy use of Sino-Vietnamese words, which directly impact young students’ comprehension but are rarely exploited in existing summarization systems.

This study proposes MAS-ViSum, a multi-agent Vietnamese text summarization framework with explicit difficulty control tailored for primary school students (Grades 1–5). The system integrates extractive and abstractive summarization with a readability adjustment mechanism based on linguistic features and a grade-stratified vocabulary dictionary, generating concise, coherent, and pedagogically suitable summaries. Contributions of this paper are threefold:

- A multi-agent framework called MAS-ViSum that integrates two specialized models — VES (PhoBERT-based extractive summarization) and VAS (ViT5-based abstractive summarization) — with dynamic grade-aware difficulty control.
- A dual-level difficulty adaptation mechanism supported by a large educational dataset and grade-specific Vietnamese vocabulary dictionary.
- An evaluation that demonstrates competitive performance on automatic metrics along with improved readability and educational suitability for primary school students, as assessed through preliminary human evaluation by an experienced teacher.

II. RELATED WORK

Text summarization is a fundamental NLP task that produces concise document versions while preserving key information [3, 4], broadly categorized into extractive methods, which select salient sentences from the source, and abstractive methods, which generate novel sentences to convey main ideas more naturally [7]. While both approaches have proven effective in improving reading comprehension and logical thinking [1, 2], most systems target general domains such as news or scientific articles and lack mechanisms to adapt linguistic difficulty for primary school students, particularly in low-resource languages like Vietnamese [8, 9].

LLMs have significantly advanced summarization through powerful semantic understanding and generation [3, 4]. In Vietnamese NLP, PhoBERT [7] and ViT5 [8] demonstrate strong extractive and abstractive performance respectively, yet both are pretrained on general-domain corpora, limiting their effectiveness in primary education contexts requiring strict vocabulary control and pedagogical appropriateness [9, 10]. Controllable summarization techniques — including length, style, and readability steering — offer a promising direction [3, 4], but most work remains confined to high-resource languages with limited integration of grade-aware difficulty control [9, 10].

MAS provide a modular and scalable paradigm for complex NLP tasks through specialized agent coordination [5, 11], with frameworks such as LangGraph enabling dynamic orchestration and iterative refinement. However, existing MAS-based summarization systems rarely incorporate educational constraints such as grade-specific readability, particularly for low-resource languages. These gaps motivate MAS-ViSum, a multi-agent framework that unifies extractive and abstractive summarization with explicit dual-level difficulty control for Vietnamese primary education.

III. PROPOSED METHOD

A. System Overview

This paper presents MAS-ViSum, a multi-agent framework for adaptive Vietnamese text summarization tailored specifically for primary school students (Grades 1–5). The framework addresses the challenge of generating summaries that are not only concise and semantically accurate but also pedagogically appropriate to the cognitive and linguistic abilities of young learners. The proposed system operates through three main stages: input preprocessing for normalization and context preparation, LangGraph-based multi-agent coordination of extractive and abstractive summarization, and a dual-level difficulty control mechanism that dynamically adjusts linguistic complexity according to both the target grade level and the desired difficulty mode (Easy, Medium, or Hard). By tightly integrating these capabilities within a collaborative multi-agent architecture, MAS-ViSum

enables flexible content generation, iterative quality refinement, and fine-grained readability control, distinguishing it from conventional “one-size-fits-all” summarization approaches and making it particularly suitable for educational applications in low-resource languages such as Vietnamese. As illustrated in Fig. 1, the system supports end-to-end processing from user input through intent analysis, planning, agent orchestration, evaluation, and difficulty adaptation.

B. Multi-Agent System

The core of MAS-ViSum is a LangGraph-based multi-agent system that coordinates multiple specialized agents to achieve adaptive and high-quality Vietnamese text summarization. This architecture decomposes the complex summarization task into distinct responsibilities, enabling greater modularity, collaboration, and iterative refinement. Table I summarizes the main agents and their corresponding functions in the proposed framework.

TABLE I. AGENT DESCRIPTIONS IN THE MAS-VISUM FRAMEWORK

Agent	Function
Intent Agent	Classifies user intent and extracts key requirements
Image2Text Agent	Performs OCR to extract text from input images
Clarification Agent	Generates clarification questions when input is ambiguous
Planning Agent	Creates an execution plan based on intent, grade level, and difficulty mode
Extractor Agent	Performs extractive summarization using fine-tuned PhoBERT
Abstracter Agent	Generates abstractive summaries using fine-tuned ViT5
Evaluation Agent	Assesses summary quality using semantic and readability metrics
Memory Agent	Manages short-term session memory and long-term experience via ChromaDB

The agents collaborate through a hierarchical orchestration mechanism consisting of fast execution and deliberative reasoning with evaluation. The central Orchestrator dynamically routes tasks among agents and triggers refinement when necessary. Algorithm 1 as in Fig. 2 details the overall orchestration routing and iterative refinement process.

Algorithm 1: ORCHESTRATOR_ROUTING_AND_REFINEMENT

Input: originalText, targetGrade, difficultyMode, conversationHistory
Output: finalSummary

BEGIN

```

1: intent ← INTENT_AGENT(originalText,
                           conversationHistory)
2: IF input contains image THEN
3:   originalText ← IMAGE2TEXT_AGENT(image)
4: END IF
5: IF clarification is needed THEN
6:   question ← CLARIFICATION_AGENT(intent)
7:   RETURN question
8: END IF
9: plan ← PLANNING_AGENT(intent, targetGrade,
                        difficultyMode)
10: currentSummary ← ORCHESTRATOR(plan,
                                originalText)
11: evaluation ← EVALUATION_AGENT(currentSummary,
                                targetGrade, difficultyMode)
12: iterations ← 0

```

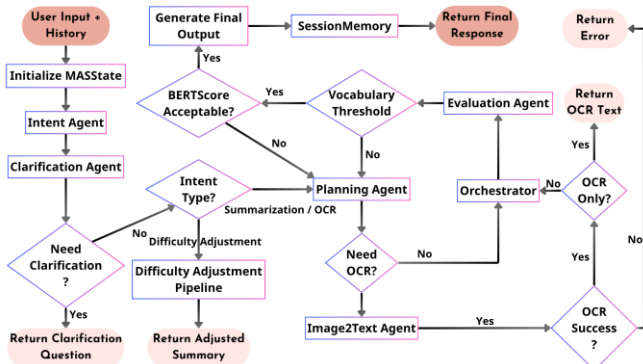


Fig. 1 Overall architecture of the proposed MAS-ViSum framework. The system integrates multiple specialized agents coordinated by an Orchestrator, with dynamic grade-aware difficulty control and iterative refinement.

```

13: WHILE (evaluation.is_acceptable() = FALSE)
    AND (iterations < MAX_ITER) DO
14:   feedback ← evaluation.get_feedback()
15:   plan ← PLANNING_AGENT.revise(plan, feedback)
16:   currentSummary ← ORCHESTRATOR(plan,
                                originalText)
17:   evaluation ← EVALUATION_AGENT(
    currentSummary, targetGrade, difficultyMode)
18:   iterations ← iterations + 1
19: END WHILE
20: STORE_CONTEXT(currentSummary, evaluation,
    conversationHistory)
21: RETURN currentSummary
END

```

Fig. 2 Pseudo-code of the Multi-Agent Orchestration and Iterative Refinement Algorithm

C. Difficulty Control Mechanism

1) Motivation and Problem Definition

Traditional text summarization systems typically focus on surface-level control such as length or compression ratio, while paying limited attention to linguistic difficulty and pedagogical suitability. This limitation is particularly critical in primary education, where summaries must match students' reading ability, vocabulary knowledge, and cognitive development. In Vietnamese, additional challenges arise from sentence structure flexibility, tonal marks, and frequent use of Sino-Vietnamese words. To address these issues, we propose a dual-level difficulty control mechanism that dynamically adjusts summary complexity according to both grade level and user-specified difficulty mode.

2) Difficulty Representation

We define difficulty using two orthogonal dimensions:

- Grade level (Grades 1 to 5), aligned with the official Vietnamese primary school curriculum.
- Difficulty mode (Easy, Medium, Hard) within each grade.

The dual-level representation allows flexible personalization while maintaining curriculum compatibility.

3) Linguistic Feature Design

To quantify readability, the system extracts a comprehensive set of linguistic features at both lexical and syntactic levels. Key features include vocabulary coverage relative to grade-specific dictionaries, rare and unknown word ratios, average word grade, average sentence length, long sentence ratio, and Sino-Vietnamese word density. These features are computed using the grade-stratified Vietnamese vocabulary dictionaries developed in this study.

4) Readability Scoring

The overall readability of a summary is measured by a weighted composite score:

$$RS(g, d) = w_1 * V_g + w_2 * (1 - R_r) + w_3 * (1 - S_{norm}) + w_4 * H_v \quad (1)$$

where:

- V_g is the grade-specific vocabulary coverage ratio,
- R_r is the rare/unknown word ratio,
- S_{norm} is the normalized average sentence length,

- H_v is the Sino-Vietnamese word ratio,
- w_1, w_2, w_3, w_4 are the weighting coefficients.

The weights were empirically tuned based on preliminary experiments and feedback from primary school teachers, resulting in $w_1 = 0.40$, $w_2 = 0.25$, $w_3 = 0.20$, and $w_4 = 0.15$. This configuration assigns the highest priority to vocabulary suitability, which is the most critical factor for young learners' comprehension. The final difficulty level (Easy, Medium, or Hard) is determined by mapping the readability score to predefined thresholds for each grade.

5) Control Strategy

The system employs different control strategies for extractive and abstractive summarization. For extractive summarization, difficulty is primarily adjusted by varying the number of selected sentences. For abstractive summarization, a post-generation refinement pipeline is applied. Algorithm 2 as in Fig. 3 details this difficulty-controlled adjustment process, which includes vocabulary simplification or controlled enrichment, sentence shortening, and semantic fidelity preservation. The difficulty controller is tightly integrated with the Evaluation Agent, allowing iterative adjustment until both semantic quality and readability requirements are satisfied.

Algorithm 2: DIFFICULTY_CONTROLLED_ADJUSTMENT
Input: draftSummary, targetGrade, targetDiff Output: adjustedSummary
BEGIN 1: current ← draftSummary 2: feats ← EXTRACT_FEATURES(current) 3: score ← COMPUTE_SCORE(feats, targetGrade) 4: iter ← 0 5: WHILE (NOT MATCH(score, targetDiff)) AND (iter < MAX_ITER) DO 6: IF score > UPPER_THRESHOLD(targetDiff) THEN 7: current ← SIMPLIFY(current, targetGrade) 8: ELSE 9: current ← ENRICH(current, targetGrade) 10: END IF 11: feats ← EXTRACT_FEATURES(current) 12: score ← COMPUTE_SCORE(feats, targetGrade) 13: iter ← iter + 1 14: END WHILE 15: RETURN current END

Fig. 3 Pseudo-code of the Difficulty-Controlled Summarization Algorithm

This mechanism ensures that every summary generated by MAS-ViSum is not only semantically accurate but also educationally appropriate for Vietnamese primary school students.

D. Summarization Models

To support both extractive and abstractive summarization within the multi-agent framework, we fine-tuned two Vietnamese pretrained language models tailored to the primary education domain.

1) Extractive Summarization Model

For extractive summarization, we fine-tuned PhoBERT [7] as a sentence importance classifier. Each sentence, together with its surrounding context, is fed into the model to predict whether it should be included in the summary. The model was trained on more than 10,000 educational documents, yielding over 240,000 labeled sentence samples. Training was optimized using the macro-averaged F1 score, with data partitioned at the document level to avoid leakage. Sentence selection combines graph-based ranking methods with PhoBERT’s semantic scoring and the Maximal Marginal Relevance (MMR) criterion to balance relevance and diversity:

$$MMR(s_i) = \lambda \cdot \text{Score}(s_i) - (1 - \lambda) \max_{s_j \in R} (E(s_i)^T E(s_j)) \quad (2)$$

where $E(\cdot)$ denotes the embedding vector produced by PhoBERT, $\text{Score}(s_i, D)$ is the relevance score between sentence s_i and the full document D , and R is the set of already selected sentences. The parameter λ controls the trade-off between relevance and redundancy. The final extractive summary is formed by concatenating the selected sentences in their original order.

2) Abstractive Summarization Model

For abstractive summarization, we fine-tuned ViT5 [8] on more than 22,000 high-quality (input, summary) pairs from educational texts. In addition to the standard negative log-likelihood loss, we incorporated ROUGE-L and BERTScore into the training objective to improve semantic fidelity and fluency. Hyper-parameter search was conducted using Optuna. The best configuration used a learning rate of 1.48×10^{-4} , batch size of 8, and 5 epochs. Techniques such as mixed precision training, gradient accumulation, warmup scheduling, and early stopping were employed to enhance training stability and efficiency.

3) Integration with the Multi-Agent Framework

The fine-tuned PhoBERT and ViT5 models are integrated into the MAS architecture through the Extractor Agent and Abstracter Agent, respectively. The Orchestrator dynamically selects the appropriate summarization approach based on the plan generated by the Planning Agent. The draft summary produced by either agent is subsequently passed to the Evaluation Agent and Difficulty Control Mechanism for quality assessment and readability refinement before being returned as the final output. This dual-model design allows MAS-ViSum to combine the strengths of extractive and abstractive paradigms while ensuring strong alignment with educational requirements.

IV. EXPERIMENTS

A. Dataset and Preprocessing

To train and evaluate the proposed models, we constructed a large-scale educational dataset specifically for Vietnamese primary school students. The corpus was collected from authentic and high-quality sources aligned with the national curriculum, including official textbooks, children’s literature,

folk tales, and informative texts. The main data sources and their statistics are summarized in Table II.

TABLE II. DATASET SOURCES AND DISTRIBUTION

Data Source	Type	Count	Publisher
Canh Dieu Textbooks	Readings, Short stories	652	Vietnam National University of Education Press
Ket Noi Tri Thuc Textbooks	Readings, Short stories	987	Vietnam Education Publishing House
Chan Troi Sang Tao Textbooks	Readings, Short stories	1,267	Vietnam Education Publishing House
Vietnamese Fairy Tales	Folk Tales	1,500	Kim Dong Publishing House
Vietnamese & World Fables	Fables	1,500	Youth Publishing House
Vietnam Children’s Literature	Modern Tales	1,500	Kim Dong Publishing House
Vietnamese History	Informative Texts	2,720	Social Sciences Publishing House

To support the difficulty control mechanism, we also constructed grade-stratified Vietnamese vocabulary dictionaries. Using VnCoreNLP for word segmentation and normalization, we built cumulative vocabularies for Grades 1 to 5, where each higher grade inherits all words from previous grades while adding new ones. The vocabulary sizes are shown in Table III.

TABLE III. GRADE-LEVEL VIETNAMESE VOCABULARY DICTIONARY

Grade	Dictionary Name	Vocabulary Size
1	grade_1.txt	3,102
2	grade_2.txt	5,314
3	grade_3.txt	7,689
4	grade_4.txt	10,959
5	grade_5.txt	13,677

For extractive summarization, documents were sentence-segmented and labeled, producing more than 240,000 training samples. Each sentence was then ranked using a unified extractive scoring function that combines graph-based importance, semantic relevance, and redundancy reduction to determine the most informative sentences for the final summary. For abstractive summarization, we generated high-quality summary pairs, resulting in over 22,000 samples. This process was formulated as a constrained optimization problem that jointly maximizes grade-level vocabulary compatibility and semantic preservation while controlling summary length. All data splits were performed at the document level (80/10/10 for train/validation/test) to ensure no information leakage between splits. Text preprocessing steps included cleaning, normalization, and removal of noisy content to maintain high data quality.

B. Experimental Setup

All experiments were conducted on GPU-accelerated computing environments. Initial development and hyper-parameter tuning were performed on Kaggle notebooks. The main training runs were executed on a cloud platform (Vast.ai) using NVIDIA RTX 3090 GPUs.

TABLE IV. HYPER-PARAMETERS FOR FINE-TUNING MODELS

Model	Task	Learning Rate	Batch Size	Epochs	Optimization Method
PhoBERT	Extractive	2.24×10^{-5}	16	6	Optuna, F1 Macro
ViT5	Abstractive	1.48×10^{-4}	8	5	Optuna, ROUGE-L, BERTScore

The multi-agent system was implemented using the LangGraph framework for agent orchestration and state management. Conversation history and long-term experience were stored and retrieved using the ChromaDB vector database. During inference, the Evaluation Agent assesses each generated summary based on semantic quality (ROUGE and BERTScore) and readability (vocabulary coverage). If the summary fails to meet the predefined thresholds (e.g., vocabulary coverage ratio below 0.8), the system triggers up to three iterative refinement cycles. For abstractive summarization, three output length modes were supported: short (~30%), medium (~40%), and long (~60%) of the original document length. Evaluation was performed on a held-out test set of 4,983 documents from VietJack to ensure fair comparison with baseline models.

C. Baselines

This study conducts a comparative analysis against several widely adopted text summarization methods. Both models are evaluated and compared against the baselines reported in Table V, with the aim of assessing their capacity for salient content selection, summary generation quality, fluency of expression, and suitability for primary school students

TABLE V. BASELINE METHODS AND MODELS

Category	Method	Type
Extractive	Lead-N	Heuristic
Extractive	TextRank	Graph-based
Extractive	LexRank	Graph-based
Extractive	BERTSUM	Transformer-based
Abstractive	Qwen2.5-3B-Instruct	LLM
Abstractive	Llama-3.2-3B-Instruct	LLM
Abstractive	GPT-4o Mini	LLM
Proposed	VES	Proposed Extractive Model
Proposed	VAS	Proposed Abstractive Model

D. Evaluation Metrics

This study employs ROUGE-1, ROUGE-2, and ROUGE-L to evaluate the degree of information preservation between the generated summaries and the reference texts. In addition, BERTScore was employed to assess semantic similarity based on contextual embeddings from Transformer-based models, which is particularly suitable for evaluating abstractive summarization. Readability was further quantified through a comprehensive set of linguistic features, including total words, unique words, rare word ratio, unknown word ratio, average word grade, number of sentences, average sentence length, long sentence ratio, average word length, words per sentence, and Sino-Vietnamese word count.

E. Results

1) Automatic Evaluation Results

Based on Table VI, the VAS model achieves a BERTScore of 0.76 and ROUGE-1 of 0.82, significantly outperforming Qwen and LLaMA in semantic preservation and retention of key information. ROUGE-2 (0.55) and ROUGE-L (0.56) remain competitive despite not surpassing GPT. It should be noted that ROUGE and BERTScore do not fully capture the linguistic appropriateness of VAS, which is specifically designed with readability control objectives — including restricted vocabulary usage, prioritization of simple sentence structures, and reduced reliance on difficult words.

TABLE VI. EVALUATION RESULTS ON BASELINE METHODS

Method / Models	Type Summary	ROUGE-1	ROUGE-2	ROUGE-L	BERT Score
Lead-n	Extractive	0.78	0.53	0.52	0.73
TextRank	Extractive	0.83	0.65	0.66	0.79
LexRank	Extractive	0.82	0.62	0.63	0.78
BertSum	Extractive	0.81	0.59	0.61	0.75
VES	Extractive	0.77	0.52	0.53	0.71
Qwen2.5-3B	Abstractive	0.71	0.45	0.47	0.71
Llama3.2-3B	Abstractive	0.68	0.43	0.45	0.69
GPT-4o mini	Abstractive	0.86	0.64	0.66	0.80
VAS	Abstractive	0.82	0.55	0.56	0.76

VES yields lower scores compared to the baselines, primarily because the concise structure and well-defined main ideas of the dataset allow graph-based methods to more readily identify central sentences. Furthermore, the use of GPT-generated reference summaries introduces an extractive-oriented bias that advantages traditional approaches. Nevertheless, VES maintains stable performance and demonstrates considerable potential when integrated with the readability control mechanism within the multi-agent system.

2) Readability Evaluation Results

TABLE VII. READABILITY EVALUATION OF SUMMARIZATION METHODS

Indicators	References	Qwen2.5-3B	Llama3.2-3B	GPT-4o mini	Our Model
total_words	152.206	102.828	89.543	136.813	148.028
unique_words	98.591	69.910	59.772	92.552	97.124
rare_word_ratio	0.057	0.078	0.066	0.062	0.060
unknown_word_ratio	0.050	0.067	0.057	0.053	0.052
avg_word_grade	1.158	1.211	1.187	1.182	1.167
num_sentences	9.996	7.374	6.247	7.720	8.698
avg_sentence_length	14.480	14.108	14.534	16.894	16.423
long_sentence_ratio	0.458	0.392	0.402	0.671	0.630
avg_word_length	3.728	3.760	3.769	3.830	3.751
words_per_sentence	15.384	14.874	15.386	17.798	17.335
hanviet_word_count	25.921	17.042	14.671	22.993	25.660

The results as in Table VII indicate that all models generate summaries shorter than the reference text, with Qwen and LLaMA producing the lowest word counts but exhibiting higher rare word ratios (0.078) and unknown word ratios (0.067). Regarding lexical difficulty, the proposed model achieves the lowest avg_word_grade of 1.167, closely approximating the reference text (1.158). The model further generates an average of 8.70 sentences with 16.42 words per sentence and a long_sentence_ratio of 0.630, while maintaining a Sino-Vietnamese word count closely aligned with the

reference text (25.66 versus 25.92). Furthermore, despite the readability adaptation mechanism not yet being applied, the proposed model already demonstrates a tendency to generate summaries whose readability characteristics more closely resemble those of the reference text compared to all baselines, indicating its superior suitability for primary school students.

V. DISCUSSION

Experimental results demonstrate that the proposed MAS-ViSum framework effectively improves summarization quality for Vietnamese primary education. The system achieves competitive performance on ROUGE and BERTScore while delivering better readability and educational suitability compared to conventional baselines. In extractive summarization, MAS-ViSum maintains stable performance in selecting educationally relevant sentences and excels in readability control and multi-agent integration, despite graph-based methods achieving higher ROUGE scores. For abstractive summarization, it attains a ROUGE-1 of 0.82 and BERTScore of 0.76, outperforming Qwen2.5-3B and Llama-3.2-3B, with vocabulary difficulty (avg_word_grade = 1.167) closely aligned to reference texts (1.158). A preliminary human evaluation was conducted with an experienced primary school teacher with over 25 years of teaching experience. The teacher indicated that the generated summaries are generally clear, coherent, and useful for classroom learning, though occasional guidance may still be needed for younger students. Architecturally, the multi-agent design provides clear role separation, iterative refinement, and scalability, surpassing traditional single-model pipelines. The study also contributes a specialized educational dataset and grade-stratified vocabulary dictionary for Vietnamese. Limitations include processing latency for long documents and the relatively modest scale of the training data, which will be addressed in future work.

VI. CONCLUSIONS

This paper introduced MAS-ViSum, a multi-agent framework for Vietnamese text summarization with grade-aware dual-level difficulty control tailored for primary school students (Grades 1–5). By integrating fine-tuned PhoBERT and ViT5 models with linguistic readability scoring and iterative agent collaboration, the system generates semantically accurate and pedagogically appropriate summaries. Experimental results show competitive performance on automatic metrics alongside improved readability and educational suitability. The study also contributes a specialized educational dataset and grade-stratified vocabulary dictionary for Vietnamese NLP. Future work will address inference latency, dataset scaling, and personalized adaptation. MAS-ViSum represents a promising step toward intelligent educational tools in low-resource languages.

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